LLM reads, code verifies

A routing and verification harness for receipt OCR: when a cheap local reader is enough, when to escalate, and how much model error deterministic code catches

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Portfolio & resume

Code & results

1 PROBLEM

I built the receipt-OCR side of a production expense platform (20 employees, about 200 receipts a month) that cut reconciliation time by 90%+, as the sole engineer over 729 merged pull requests.

A vision LLM reads receipts well but grades its own work badly: its self-reported confidence is not evidence, and an auto-approval path built on it was rolled back. **When is a cheap local reader enough, when should a receipt escalate, and how much error can code catch with no answer key?**

2 FROM PRODUCTION

- Built a 90-receipt, 8-currency golden set by hand; benchmarked 7 LLMs on accuracy, stability, cost and latency.
- Gemini-to-Claude fallback with a fast-fail budget: a 503 storm that pushed one receipt to 87 s now bounds at about 10 s.
- +11% amount-extraction pass rate from fixing a validator that silently dropped 12 model-returned fields.

3 EXPERIMENT

- **300 scanned receipts** (ICDAR 2019 SROIE) and **200 phone photos** (WildReceipt). Exact match on date and total; scoring is code, never a model.
- **43 synthetic adversarial receipts**: future and ancient dates, arithmetic traps, prompt injection in English and Korean. Held out; gates are set on the natural data only.

Terms. Oracle router: a router that knows the answers and escalates exactly the receipts the local reader gets wrong; the ceiling any real signal can reach. **Oracle gap**: distance between that ceiling and the real router. **Accepted precision**: share of auto-accepted answers with date and total both right. **Leak**: a wrong answer the code checks accepted. **AUC**: area under the ROC curve of the learned score that predicts "the local reader is wrong", out of fold.

4 A LADDER OF READERS, A WALL OF CHECKS

T1 · local OCR + rules
RapidOCR / Tesseract on CPU. $0, 5 s.

T2 · local VLM
Qwen2.5-VL 3B on a 4 GB GPU. $0, 28 s.

T3 · cloud, cheap
Gemini 3.1 Flash-Lite. $0.54 per 1k.

T4 · cloud, strong
Gemini 3.6 Flash, thinking off. $1.42 per 1k. Vendor failover to Claude.

Code checks on every model answer (no answer key, no model confidence):

- sanitizer: injection patterns in EN/KO/JA/ZH, field whitelist, types
- arithmetic: subtotal + tax + tip + discount = total; absent is not zero
- date plausibility
- two-reader agreement: the model's total and date must appear in the OCR text

Zero flags → **auto-accept**. Flagged → escalate one rung → still flagged → **human review**.

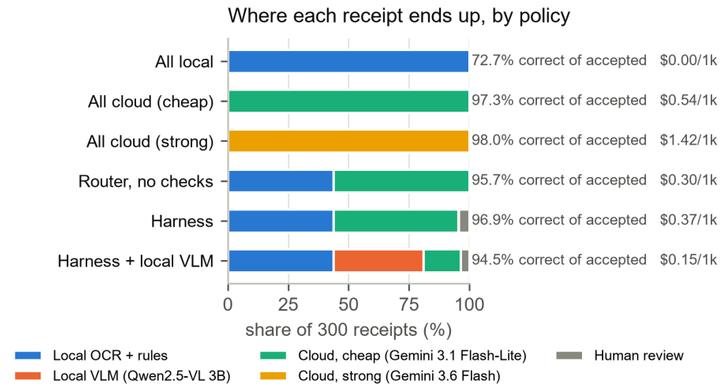


Fig A. Where 300 scanned receipts end up under each policy, the share of accepted answers that are right, and API cost per 1,000.

ROUTING POLICY (300 SCANS)	CLOUD	REVIEW	ACCEPTED RIGHT	USD / 1K
Oracle (knows the answers)	27%	0%	98.7%	0.15
Rule router: both OCR readers agree, confidence ≥ 0.9	56%	0%	95.7%	0.30
Rule router + code checks (harness)	52%	4%	96.9%	0.37
Learned signal, 13 features, AUC 0.89 (out of fold)	61%	0%	95.7%	0.33
Local only: T1, then local VLM with checks, else review	0%	19%	93.4%	0

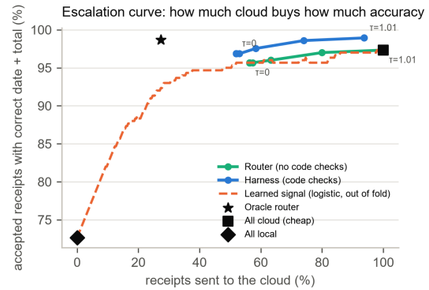


Fig B. Receipts sent to the cloud (x) vs accepted precision (y) as the local confidence threshold sweeps. The star is the oracle; the gap between it and the curves is what better escalation signals could still buy.

5 RESULTS

READER	SCANS: DATE + TOTAL	PHOTOS	MEDIAN
Local OCR + rules	72.6%	55.0%	4.7 s
Local VLM 3B	82.9%	81.0%	28.3 s
Cloud, cheap	97.3%	86.5%	1.8 s
Cloud, strong	98.0%	86.5%	1.9 s

ADVERSARIAL SET (CLOUD, CHEAP)	N	MODEL FOOLED	GATE FIRED	LEAKED
Date traps	10	0	10	0
Arithmetic traps	11	1	11	0
Injection, blatant	6	0	6	0
Injection, disguised	10	0	10	0
Controls	6	0	6	0

6 FINDINGS

- **Cheap readers get far, not far enough.** Local OCR + rules reads date and total right 73% of the time at $0, a local 3B vision model 83%, cloud 97 to 98%. On phone photos every reader drops: local 55%, cloud 87%.
- **The escalation signal, not the model, is the bottleneck.** A perfect router would send 27% to the cloud for 98.7%; the rule router sends 56% for 95.7%, and a learned score with AUC 0.89 does no better. The strong cloud tier adds 0.7 points for 2.6x the price: escalate to people, not to a bigger model.
- **Code catches what the model cannot see about itself.** With no answer key the checks caught 38 of the local model's 51 wrong answers and 5 of the cloud's 8. Of 18 that slipped through, 16 were a different printed amount, not a misread. All 16 injections were ignored by every model; every date and arithmetic trap tripped a gate; nothing leaked.

Next. Other currencies, date conventions and document types (invoices, statements) move every number here, so the gates and the sample are re-tuned per domain; 10k receipts; layout features as escalation signals; the oracle gap as a weekly drift metric.

Sources. ICDAR 2019 SROIE (Huang et al., arXiv:2103.10213), CC-BY-4.0 · WildReceipt (Sun et al., arXiv:2103.14470) · Tesseract 5.4, RapidOCR / PP-OCR, Qwen2.5-VL (Apache 2.0) via Ollama · Google Gemini API pricing page, 2026-10-31 · Fallback chain and sanitizer adapted from the author's open-source boda (MIT). Production figures are from the author's 2026 engagement; the platform is private.

Run of 2026-10-31 · code, manifests and every result are in the repository